

EX (BINOMIAL MM, CONTINUED)

METHOD OF MOMENTS

FROM LAST LECTURE

INDEPENDENT
IDENTICALLY
DISTRIBUTEDUNKNOWN N AND P , A IID SAMPLE OF m X_i

(... LAST TIME ...)

MM ESTIMATOR FOR P

$$\begin{cases} \tilde{p} = \bar{X} / \tilde{N} \leftarrow \text{ESTIMATE of } N \\ \tilde{N} = \frac{\bar{X}^2}{\bar{X} - \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2} \end{cases}$$

THEN WE REALIZED THAT IT CAN WORK BADLY WHEN m SMALL

$$\begin{aligned} & \frac{1}{m} \sum_{i=1}^m (X_i^2 + \bar{X}^2 - 2\bar{X}X_i) \\ &= \frac{1}{m} \sum_{i=1}^m X_i^2 + \frac{1}{m} \sum_{i=1}^m \bar{X}^2 - 2\bar{X} \frac{1}{m} \sum_{i=1}^m X_i \\ &= \bar{X}_2 + \bar{X}^2 - 2\bar{X} \cdot \bar{X} = \bar{X}_2 - \bar{X}^2 \end{aligned}$$

$$\tilde{N} = \frac{\bar{X}^2}{\bar{X} - (\bar{X}_2 - \bar{X}^2)}$$

!!! IN THE EXAM

MAXIMUM LIKELIHOOD METHOD (MLE)

WIDELY USED BECAUSE OF ITS NICE PROPERTIES

YOU FIRST FIND POSSIBLE VALUES FOR θ , CHOOSE THAT ONE FOR WHICH THE DATA YOU ACTUALLY OBSERVE HAS THE MAXIMUM PROBABILITY TO BE OBSERVEDIDEA: FIND THAT VALUE OF θ AT WHICH PROB. OF DATA IS MAXIMUMEXTREME EX (EXAMPLE) $\underline{x} = (0.001, 0.003, -0.003)$ COMPARE TWO
NORMAL DISTRIBUTION
WHICH NORMAL DISTRIBUTION
SEEMS MORE REASONABLE
FOR THE DATA YOU HAVE?

$N(\mu=0, \sigma^2=1)$ MORE COHERENT TO $\underline{x}$

$N(\mu=10, \sigma^2=1)$ LESS COHERENT

BECAUSE THE OBSERVATIONS
ARE CLOSER TO 0 THAN
TO 10MAXIMUM
LIKELIHOOD
ESTIMATION
 $\Rightarrow$ MLE SUGGESTS
TO PREFER
 $\mu=0$

FOR $\underline{x} = (x_1, x_2, \dots, x_m)$ REALIZATION OF IID SAMPLE
FROM SOME POP. $f(x; \theta)$

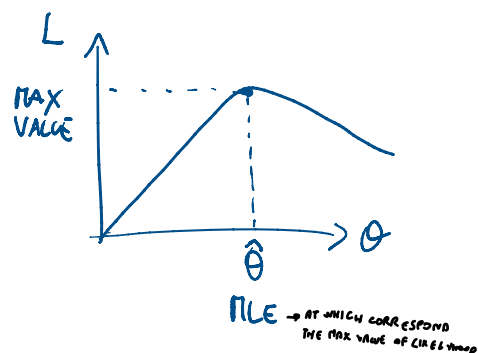
WE DEFINE THE LIKELIHOOD FUNCTION AS:

SAMPLE SIZE ASSUMED TO BE FIXED

$$L(\theta; \underline{x}) = \prod_{i=1}^m f(x_i; \theta)$$

DIFFERENCE: FUNCTION OF θ , NOT OF $\underline{x}$

JOINT PDF/PMF OF $\underline{x}$



THAT IS: THE PROBAB. (OR DENSITY) OF THE DATA, AS A FUNCTION OF THE PARAMETER

IF $L(\theta^{(1)}; \underline{x}) > L(\theta^{(2)}; \underline{x})$, THEN WE PREFER $\theta^{(1)}$.

$$\Rightarrow \hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(\theta; \underline{x})$$

MAXIMUM LIKELIHOOD ESTIMATOR / ESTIMATE OF θ

θ AT WHICH L IS MAX

IN PRACTICE: SIMPLER TO MAXIMIZE THE LOG-LIKELIHOOD:

$$\begin{aligned} \ell(\theta; \underline{x}) &= \ln L(\theta; \underline{x}) = \ln \prod_{i=1}^m f(x_i; \theta) \quad \text{NOT THE LIKELIHOOD} \\ &= \sum_{i=1}^m \ln f(x_i; \theta) \end{aligned}$$

NATURAL LOGARITHM

(IT WILL GIVE THE SAME $\hat{\theta}$, SINCE LOG MONOTONE INCREASING TRANSFORMATION)

AND TO FIND $\hat{\theta}$: $\frac{\partial}{\partial \theta} \ell(\theta; \underline{x}) \stackrel{\text{FIX}}{=} 0 \Rightarrow$ FIND θ SOLUTION

. THEN CHECK THAT $\frac{\partial^2}{\partial^2} \ell(\theta; \underline{x}) < 0$ (TO BE A MAX)

• THEN CHECK THAT $\left. \frac{\partial^2}{\partial \theta^2} \ell(\theta; \underline{x}) \right|_{\theta = \hat{\theta}} < 0$ (TO HAVE A MAX)

• YOU FINALLY CHECK VALUES OF $\ell(\theta; \underline{x})$ AT THE BOUNDARIES OF (M)

EX: $X \sim \text{Bern}(p)$. $f(x; p) = p^x (1-p)^{1-x}$, $x \in \{0, 1\}$

$$\begin{aligned} \underline{L(p; \underline{x})} &= \prod_{i=1}^m f(x_i; p) = \prod_{i=1}^m p^{x_i} (1-p)^{1-x_i} \\ &= p^{\sum_{i=1}^m x_i} (1-p)^{m - \sum_{i=1}^m x_i} \end{aligned}$$

$$\begin{aligned} \Rightarrow \underline{\ell(p; \underline{x})} &= \ln L(p; \underline{x}) = \ln \left(p^{\sum x_i} (1-p)^{m - \sum x_i} \right) \\ &= \ln p^{\sum x_i} + \ln (1-p)^{m - \sum x_i} \\ &= \sum_i x_i \cdot \ln p + (m - \sum_i x_i) \cdot \ln (1-p) \end{aligned}$$

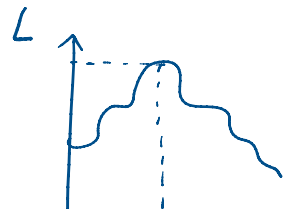
ALTERNATIVE WAY: $f(x; \theta) \rightarrow \ln f(x; \theta) \rightarrow \ell = \sum_i \ln f$

$$\begin{aligned} \hookrightarrow \ln f(x; \theta) &= \ln (p^x (1-p)^{1-x}) = x \cdot \ln p + (1-x) \ln (1-p) \\ \ell(p; \underline{x}) &= \sum_{i=1}^m \ln f(x_i; p) = \sum_{i=1}^m (x_i \ln p + (1-x_i) \ln (1-p)) \\ &= \sum_i x_i \cdot \ln p + (m - \sum_i x_i) \ln (1-p) \end{aligned}$$

$$\frac{\partial}{\partial p} \ell(p; \underline{x}) = \frac{\sum_i x_i}{p} + \frac{m - \sum_i x_i}{1-p} \cdot (-1) \stackrel{\text{fix}}{=} 0$$

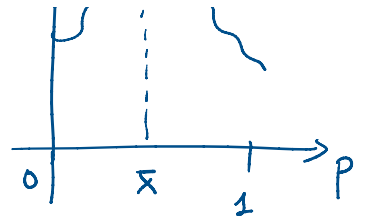
$$\Rightarrow \frac{m \bar{x}}{\hat{p}} - \frac{m - m \bar{x}}{1 - \hat{p}} = 0 \quad \Rightarrow \quad \frac{\bar{x}}{\hat{p}} - \frac{1 - \bar{x}}{1 - \hat{p}} = 0$$

$$\Rightarrow \bar{x} - \hat{p} \bar{x} - \hat{p} + \hat{p} \bar{x} = 0 \quad \Rightarrow \quad \hat{p} = \bar{x}$$



$$\frac{\partial^2}{\partial p^2} \ell(p; \underline{x}) = \frac{\partial}{\partial p} \left(\frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} \right)$$

$$= -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1-p)^2} < 0$$



$$\Rightarrow \hat{p} = \bar{x} \text{ MLE}$$

$$\lim_{p \rightarrow 0} \ell(p; \underline{x}) = -\infty, \quad \lim_{p \rightarrow 1} \ell(p; \underline{x}) = -\infty$$

EX: $X \sim \text{Pois}(\lambda)$, $f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x = 0, 1, 2, \dots$

$$\ln f(x; \lambda) = \ln \left(\frac{\lambda^x e^{-\lambda}}{x!} \right) = \ln \lambda^x + \ln e^{-\lambda} - \ln x!$$

$$= x \cdot \ln \lambda - \lambda - \ln x!$$

$$\ell(\lambda; \underline{x}) = \sum_{i=1}^n \ln f(x_i; \lambda) = \sum_{i=1}^n (x_i \ln \lambda - \lambda - \ln x_i!)$$

$$= \ln \lambda \cdot \sum_{i=1}^n x_i - n\lambda - \sum_{i=1}^n \ln x_i!$$

$$\frac{\partial}{\partial \lambda} \ell(\lambda; \underline{x}) = \frac{\sum x_i}{\lambda} - n = 0 \stackrel{\text{fix}}{=} 0 \Rightarrow \frac{\sum x_i}{\hat{\lambda}} - n = 0$$

$$\Rightarrow \hat{\lambda} = \bar{x}$$

$$\frac{\partial^2}{\partial \lambda^2} \ell(\lambda; \underline{x}) = -\frac{\sum x_i}{\lambda^2} < 0$$

AT THE BOUNDARIES OF (4): $\lim_{\lambda \rightarrow 0} \ell(\lambda; \underline{x}) = -\infty$

$$\lim_{\lambda \rightarrow \infty} \ell(\lambda; \underline{x}) = -\infty$$

EX: $X \sim \text{Exp}(\lambda)$

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

$$\ln f(x; \lambda) = \ln(\lambda e^{-\lambda x}) = \ln \lambda + \ln e^{-\lambda x} = \ln \lambda - \lambda x$$

$$\ell(\lambda; \underline{x}) = \sum_{i=1}^n \ln f(x_i; \lambda) = \sum_{i=1}^n (\ln \lambda - \lambda x_i) = n \ln \lambda - \lambda \sum_{i=1}^n x_i$$

$$l(\lambda; \underline{x}) = \sum_{i=1}^n f(x_i; \lambda) = \sum_{i=1}^n (\ln \lambda - \lambda x_i) = n \ln \lambda - \lambda \sum_{i=1}^n x_i$$

$$\frac{\partial}{\partial \lambda} l(\lambda; \underline{x}) = \frac{n}{\lambda} - \sum_{i=1}^n x_i \stackrel{\text{fix}}{=} 0 \Rightarrow \frac{n}{\hat{\lambda}} = \sum_{i=1}^n x_i \Rightarrow \hat{\lambda} = \frac{n}{\sum_{i=1}^n x_i} = \bar{x}^{-1}$$

$$\frac{\partial^2}{\partial \lambda^2} l(\lambda; \underline{x}) = -\frac{n}{\lambda^2} < 0, \quad \lim_{\lambda \rightarrow 0} l(\lambda; \underline{x}) = -\infty, \quad \lim_{\lambda \rightarrow \infty} l(\lambda; \underline{x}) = -\infty \Rightarrow \bar{x}^{-1} \text{ is the MLE of } \lambda$$

EX:

$$X \sim N(\mu, 1) \quad f(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x-\mu)^2\right\}, \quad x \in \mathbb{R}$$

$$\begin{aligned} \ln f(x; \mu) &= \ln\left(\frac{1}{\sqrt{2\pi}}\right) + \ln \exp\left\{-\frac{1}{2}(x-\mu)^2\right\} \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2}(x-\mu)^2 \end{aligned}$$

$$l(\mu; \underline{x}) = \sum_{i=1}^n \ln f(x_i; \mu) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial}{\partial \mu} l(\mu; \underline{x}) = 0 - \frac{1}{2} \sum_{i=1}^n 2(x_i - \mu) \cdot (-1) = \sum_{i=1}^n (x_i - \mu) \stackrel{\text{fix}}{=} 0$$

$$\Rightarrow \sum_{i=1}^n x_i - \sum_{i=1}^n \hat{\mu} = 0 \Rightarrow \sum_{i=1}^n x_i = n \hat{\mu} \Rightarrow \hat{\mu} = \bar{x}$$

$$\frac{\partial^2}{\partial \mu^2} l(\mu; \underline{x}) = \frac{\partial}{\partial \mu} \sum_{i=1}^n (x_i - \mu) = -n < 0$$

$$\lim_{\mu \rightarrow -\infty} l(\mu; \underline{x}) = \lim_{\mu \rightarrow -\infty} l(\mu; \underline{x}) = -\infty$$

$$\Rightarrow \hat{\mu} = \bar{x} \text{ is MLE of } \mu, \quad X \sim N(\mu, 1)$$

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MULTI-PARAMETER EXTENSION:

$$\theta = (\theta_1, \theta_2, \dots, \theta_k), \quad k \geq 1 \quad \text{THEN}$$

$$\frac{\partial}{\partial \theta_i} l(\theta; \underline{x}) \stackrel{\text{fix}}{=} 0 \quad \text{FOR } i = 1, \dots, k$$

$$\frac{\partial}{\partial \theta_j} \ell(\theta; \underline{x}) \stackrel{\text{Fix}}{=} 0, \text{ FOR } j=1, \dots, k$$

+ CHECK THAT THE HESSIAN MATRIX

$$H := \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ell(\theta; \underline{x}) \right)_{i,j} \text{ IS NEGATIVE-DEFINITE}$$

EX:

$$X \sim N(\mu, \sigma^2) \quad \theta = (\mu, \sigma^2)$$

$$f(x; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$

$$\ln f(x; \theta) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(x-\mu)^2$$

$$\ell(\theta; \underline{x}) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\begin{cases} \frac{\partial}{\partial \mu} \ell(\theta; \underline{x}) = 0 - \frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \stackrel{\text{Fix}}{=} 0 \end{cases}$$

$$\begin{cases} \frac{\partial}{\partial \sigma^2} \ell(\theta; \underline{x}) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \stackrel{\text{Fix}}{=} 0 \quad (*) \end{cases}$$

$$\begin{cases} \hat{\mu} = \bar{x} \quad (\text{SAME STEPS AS PREVIOUS EXAMPLE}) \\ -\frac{n}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4} \sum_{i=1}^n (x_i - \hat{\mu})^2 = 0 \end{cases}$$

$$\begin{cases} \hat{\mu} = \bar{x} \\ -\frac{n}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4} \sum_{i=1}^n (x_i - \bar{x})^2 = 0 \end{cases}$$

$$\begin{cases} \hat{\mu} = \bar{x} \\ -n + \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \Rightarrow \begin{cases} \hat{\mu} = \bar{x} \\ \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n-1}{n} \cdot s^2 \end{cases} \end{cases}$$

(*) NOTE THAT
 (σ^2) IS THE PAR
 $\downarrow$
 $=: \eta$

$$\frac{\partial}{\partial \eta} \frac{1}{2\eta} = -\frac{1}{2\eta^2}$$

$$\left\{ -n + \frac{1}{\hat{\sigma}^2} \sum_i (x_i - \bar{x})^2 \right\} \quad \left\{ \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n-1}{n} \cdot S^2 \right.$$

$$\Rightarrow \hat{\theta} = \left(\bar{x}, \frac{n-1}{n} S^2 \right)$$

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SOMETIMES: NOT POSSIBLE TO FIND AN ANALYTICAL SOLUTION TO NLE

$\Rightarrow$ NUMERICAL PROCEDURES TO MAXIMIZE $l(\theta; x)$.

FOR INSTANCE:

• NEWTON-RAPHSON ALGORITHM:

$$\theta^{(t+1)} = \theta^{(t)} + \mathcal{J}(\theta^{(t)})^{-1} \cdot S(\theta^{(t)}), \quad \text{WHERE}$$

$S(\theta)$ IS THE VECTOR OF 1ST DERIVATIVES OF $l(\theta; x)$

AND $\mathcal{J}(\theta)$ IS THE MATRIX OF 2ND DERIVATIVES OF $l(\theta; x)$
MULTIPLIED BY -1 . (OBSERVED INFORMATION MATRIX)

• FISHER-SCORING ALGORITHM:

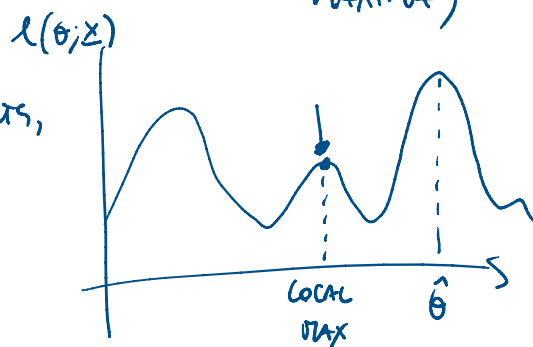
$$\theta^{(t+1)} = \theta^{(t)} + I(\theta^{(t)})^{-1} \cdot S(\theta^{(t)}), \quad \text{WHERE}$$

$$I(\theta) = E[\mathcal{J}(\theta)] \quad (\text{EXPECTED INFORMATION MATRIX})$$

WITH NUMERICAL OPTIMIZATION: NEVER SURE THAT WE REACH

TO A GLOBAL MAX. (POSSIBLE TO BE STUCK IN LOCAL MAXIMA)

TO AVOID LOCAL MAX POINTS,
RE-RUN MORE TIMES
FROM DIFFERENT
STARTING POINTS



PROPERTIES OF MLE

1) $\hat{\theta}$ depends on $\underline{x}$, only THROUGH THE SUFFICIENT STATISTIC OF θ

CONSEQUENCE OF THE FACTORIZATION TH:

$$L(\theta; \underline{x}) = f(\underline{x}; \theta) = g(\underbrace{t(\underline{x})}_{\text{SUFFICIENT}}, \theta) \cdot h(\underline{x})$$

$$l(\theta; \underline{x}) = \ln g(t(\underline{x}), \theta) + \ln h(\underline{x})$$

$$\frac{\partial}{\partial \theta} l(\theta; \underline{x}) = \frac{\partial}{\partial \theta} \ln g(t(\underline{x}), \theta) + 0 \quad \begin{array}{l} \text{DEPENDENT ON } \underline{x} \\ \text{ONLY BY } t(\underline{x}) \end{array}$$

$\Rightarrow$ IF WE HAVE $\underline{x}_1$ AND $\underline{x}_2$ FOR WHICH $t(\underline{x}_1) = t(\underline{x}_2)$ $\Rightarrow$ MLE OF θ EQUAL IN BOTH CASES

"MLE AGREES WITH SUFFICIENCY PRINCIPLE"